



The AI operations brain for retail. Agents that **show their math** — a human on every action.

ShelfWise runs a supermarket's interlocking daily decisions — expiry markdowns, procurement and multi-source stock sourcing, till-price integrity, cold-chain response — through real tool-calling agents on AMD compute. Every recommendation is grounded in computed numbers, and nothing acts without human approval.

333/333

live chat calls model-backed · zero
fallbacks

4,618

decisions created · zero ID
collisions

2,990

human approve/reject cycles ·
zero mismatches

2.0s

average live answer on MI300X

415

automated tests passing

A supermarket makes hundreds of decisions a day. Most are made late, on gut feel.

MARGIN LOST DAILY

Expiry & waste

Which stock to mark down before it expires — and by how much — is a forecasting and simulation problem, not a guess. Get it wrong and product dies on the shelf or sells below its worth.

SHELVES EMPTY, SALES WALK OUT

Procurement & sourcing

When a shortage hits, "transfer some stock" is not an answer. From which branch? The distribution centre? A supplier? At what distance and lead time — and who covers the remainder?

SILENT LEAK AT EVERY TILL

Pricing integrity

Till prices drift from catalogue prices. Every unnoticed mismatch is silent margin loss on one side — or a compliance risk on the other. Nobody is checking every SKU, every day.

A COUNTDOWN ON THOUSANDS IN STOCK

Cold chain

A fridge outage starts a clock on everything inside it. Risk must be scored against temperature, time, and product sensitivity — not eyeballed by whoever notices first.

Retail doesn't lose its profit in one catastrophe. It loses it in **a thousand small decisions made without the math** — every day, in every store.

Every answer flows through real agents, calling real tools, on AMD compute.

1 • EVENTS IN

The store streams in

POS sales, stock updates, barcode scans, cold-chain sensor alerts, and connector webhooks land on one event bus with provenance and quarantine.



2 • AGENTS REASON

Critic / Executive pairs

Gemma-4-E4B-it on AMD Instinct MI300X via vLLM runs a bounded tool-calling loop — the Critic challenges, the Executive decides.



3 • TOOLS COMPUTE

Math, not vibes

11 read-only tools do the arithmetic in tested Python — the model uses them as its calculator and cannot invent figures.



4 • HUMAN APPROVES

Nothing acts alone

Every recommendation lands as a pending decision with evidence attached. Nothing writes back without sign-off. Full audit trail.

The 11 tools: live stock · demand forecast · expiry risk · reorder policy · supplier ranking · multi-source stock sourcing · cold-chain risk · price integrity · markdown simulation · open decisions · learned thresholds

The grounding guarantee — our original contribution, enforced in code: a final answer must cite the numbers its tools actually computed. An answer citing a figure no tool produced is rejected and re-run. Never shipped. Hallucination isn't discouraged here — it's structurally impossible to ship.

One chat interface runs the whole store.

ShelfWise Console — live on AMD MI300X

We are short on SKU 4011 — where should the replacement stock come from?

- ShelfWise

Take **6 units** from **store_02_sandton** — the closest branch at **5.00 km** with the fastest delivery at **2.00 hours**, beating store_09_midrand (4.00 h). That covers only part of the shortage: place a **purchase order for the remaining 12 units**.

PENDING DECISION · AWAITING HUMAN APPROVAL

Transfer 6 units from store_02_sandton + PO for 12 units

Approve

Reject

- **A real answer from the live MI300X endpoint**

It ranked branches, the DC, and suppliers by availability, distance, and lead time — and explained why the winner won.

- **One question, four-plus live tool calls**

Approvals, stock, deliveries, sourcing — resolved in a single turn and rendered as a structured report a store manager can read.

- **Honest when the data isn't there**

If a tool has no reading, ShelfWise says so. It never fabricates a number to sound complete.

- **Multi-user, multi-tenant, isolated**

The authenticated tenant always overrides anything the model writes into tool arguments. Server-side, by construction.

Every store gets a digital twin — agents earn trust there first.

REAL STORE POS · stock · sensors · suppliers

Live signals feed a persistent, continuously evolving virtual replica of the shop.



DIGITAL TWIN A living simulation, not a script

The world generator drives the twin through realistic cycles — sales, deliveries, expiries, outages — the same substrate this submission runs on.



AGENTS REHEARSE Train, stress-test, prove

Agents learn the store's rhythms and get stress-tested against its failures before a recommendation ever touches the real shop floor.



REAL DECISIONS Human-approved write-back

Only proven, evidence-backed recommendations reach a human — and only a human lets them act.

This is how robotics ships safely — train in simulation, deploy with confidence. ShelfWise brings that discipline to retail operations. In one 15-minute continuous run, the twin generated and processed:

333

world cycles simulated

full store state each cycle

34,077

events generated

26,640 accepted through intake

4,618

decisions produced

zero duplicate IDs

34/34

learning threshold movements

every one expected & receipted

Stress-tested live **before** anything was demoed.

A 15-minute continuous simulation drove the full public API against the live AMD MI300X endpoint — every scenario type, every cascade, connectors, approvals, learning — with hard failure on any offline fallback. All artifacts are committed to the repository.

333/₃₃₃

chat calls genuinely model-backed
zero offline fallbacks · zero errors

4,618

decisions created
zero ID collisions

2,990

human approve / reject cycles
zero mismatches

2.0_s

average live answer · 2.3 s P95
multi-tool reports < 6 s end-to-end

415

automated tests passing
ruff clean · CI on every push

165

machine-verified capabilities
committed manifest · CI fails on drift

AMD compute is the engine — not a checkbox.

- **AMD Instinct MI300X** — AMD Developer Cloud droplet runs every agent decision and every chat answer.
- **vLLM 0.23 on ROCm** serving google/gemma-4-E4B-it with native Gemma tool calling — the agent loop runs on the GPU, not around it.
- **131k-token context** powers long, tool-heavy agent conversations without truncation.
- **One MI300X serves the whole store's reasoning** — ~2 s answers with headroom for many concurrent stores.

Every live response carries verifiable proof headers:

```
x-shelfwise-provider: vllm_mi300x
x-shelfwise-model: google/gemma-4-E4B-it
x-shelfwise-answer-source: model
```

The app refuses to run on anything else. In production mode, any provider other than the AMD MI300X endpoint is rejected with HTTP 503. No silent fallback. No faked success.

Honest scope: demo data is a seeded single-store world. The decision logic is general and unit-tested independently of the data; wiring live multi-branch feeds is roadmap — and we say so.

Every store position, one governed AI colleague.

WHY IT'S ORIGINAL

Grounding, enforced in code

Agents that must show their math, a digital twin per store, and human-in-the-loop governance as a first-class feature — not a disclaimer. Retail AI usually demos well and governs badly. ShelfWise inverts that.

WHY IT'S USEFUL NOW

Deployable governance

Pending decisions, evidence objects, audit trails, tenant isolation, and a learning loop that adjusts thresholds from real outcomes — the parts a real retailer's risk team asks about first.

WHAT'S NEXT

The whole chain

Live POS / WMS / ERP connector feeds, multi-branch inventory, and PDF stock sheets and period reports generated from the same grounded tools. One MI300X already has the headroom.

ShelfWise — agentic reasoning, grounded in real tools, governed by a human, running on AMD.

github.com/mrlucas679/shelfwise
AMD ACT II • Track 3: Unicorn